

ANSWER ALL QUESTIONS

USEFUL INFORMATION

h = Planck's constant = 6.626×10^{-34} J. s

e = electronic charge 1.602×10^{-19} C

R = Gas constant = 8.314 J/K.mol

ϵ_0 = 8.854×10^{-12} Fm⁻¹

N_0 = 6.023×10^{23} mol⁻¹

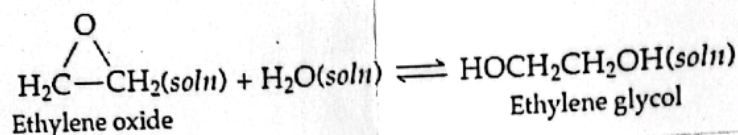
m = mass of electron 9.1079×10^{-31} kg

R_H = 109677.8 cm⁻¹ / 1.097×10^{-2} nm⁻¹.

c = speed of light = 2.998×10^8 ms⁻¹

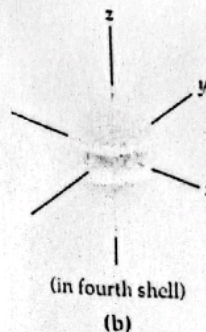
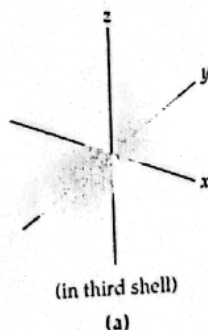
1 eV = 1.602×10^{-19} J

1. Ethylene glycol, used as antifreeze in automobile radiators, is manufactured by hydration of ethylene oxide. Write the equilibrium constant expression for K_c (3 marks)



2. Calculate the wavelength of a 57-g tennis ball travelling at a speed of 80 km/h. (Answer = 5.2×10^{-34} m; show your working) (5 marks)

3. Identify each of the following orbitals, and give n , l and m_l quantum numbers for each. (3 marks)



4. The gas-phase reaction of nitric oxide and bromine yields nitrosyl bromide: $2 \text{NO (g)} + \text{Br}_2\text{(g)} \rightarrow 2 \text{NOBr (g)}$. The rate law is $\text{rate} = k[\text{NO}]^2[\text{Br}_2]$. What is the reaction order with respect to each of the reactants, and what is the overall reaction order? (4 marks)
5. Devise a method for determination of pH of an aqueous solution (5 marks)
6. Discuss the steps involved in the construction of sp hybrid orbitals using appropriate examples. (4 marks)
7. Using equations relating A_{spec} and the area occupied by a molecule, write down an expression in the Langmuir approximation and define all terms in it. (5 marks)